

SEC & Big Ten College Football Analytics Dashboard

Updated Database Schema Documentation

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Course: MSBA-692 Data Engineering

Database Platform: Supabase PostgreSQL

1. Database Overview

The project uses a PostgreSQL database hosted on Supabase to store transformed and validated college football team records. The schema is intentionally designed as a single analytics-ready table for the MVP dashboard. This design supports efficient dashboard queries, simple maintenance, and clear alignment between the ETL pipeline and the visualization layer.

2. Final Database Table

The final database contains one primary table: **teams**. This table stores 2024 season-level team records for SEC and Big Ten programs.

Column Name	Data Type	Constraint	Description
team_name	VARCHAR(100)	PRIMARY KEY	Unique team name used as the team identifier.
conference	VARCHAR(50)	NOT NULL	Conference affiliation, filtered to SEC or Big Ten.
wins	INTEGER	NOT NULL	Total number of team wins for the season.
losses	INTEGER	NOT NULL	Total number of team losses for the season.
win_percentage	NUMERIC(5,3)	NOT NULL	Derived metric calculated from wins and losses.

3. SQL Definition

```
CREATE TABLE teams (  
    team_name VARCHAR(100) PRIMARY KEY,  
    conference VARCHAR(50) NOT NULL,  
    wins INTEGER NOT NULL,  
    losses INTEGER NOT NULL,  
    win_percentage NUMERIC(5,3) NOT NULL  
);
```

4. Primary Key and Uniqueness

The **team_name** field is used as the primary key because the final project stores one record per team for the selected 2024 season. This prevents duplicate rows from being inserted during repeated ETL executions and supports the incremental loading strategy.

5. Derived Metric

The **win_percentage** field is calculated in the ETL pipeline using the following formula:

$$\text{win_percentage} = \text{wins} / (\text{wins} + \text{losses})$$

This metric allows users to compare team performance more fairly than raw wins alone, especially when teams may have different numbers of games played.

6. Data Quality Rules

- team_name must not be null.
- conference must be either SEC or Big Ten.
- wins must be numeric and greater than or equal to zero.
- losses must be numeric and greater than or equal to zero.
- win_percentage must fall between 0 and 1.
- duplicate team records are removed before loading.

7. Example Record

team_name	conference	wins	losses	win_percentage
Georgia	SEC	12	2	0.857
Indiana	Big Ten	16	0	1.000

8. Dashboard Alignment

The schema directly supports the Dash dashboard. The **conference** field powers the interactive dropdown filter, **wins** and **losses** support the visualizations, and **win_percentage** supports the average win percentage KPI card.

9. Future Schema Expansion

Future versions of the project could expand the schema into multiple normalized tables such as conferences, teams, games, rankings, and seasons. For the MVP dashboard, the current single-table schema is appropriate because it is simple, efficient, and aligned with the available dashboard features.